Franco Andrian

Electronics Engineer | Software & Firmware Engineer

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WORK EXPERIENCE

SOFTWARE & FIRMWARE ENGINEER — **PPST Argentina** — **Jul 2023 – Present**
Software & Firmware Team — Santa Fe, Argentina

Test suite for EMC and power-quality compliance testing

- Joined at the start of the project and developed features across the full stack of its Angular + Electron client and multithreaded C++ REST server: UI, REST API, business logic, and persistence layers.
- Developed and maintained multithreaded C++ backend services for automated test execution, device management, and real-time monitoring, and designed and standardized their REST API with the Crow framework (consistent HTTP semantics, error handling, and file transfer).
- Built Angular and Angular Material user interfaces for an Electron desktop application: reusable components, RxJS-based reactive state, real-time Chart.js plots, internationalization (i18n), responsive and touch-friendly layouts, drag-and-drop tables, and a rich-text (WYSIWYG) report template editor.
- Integrated programmable power sources, DAQ devices, and proprietary instruments over USB, SCPI, TCP/IP, Serial, and mDNS.
- Implemented six IEC 61000 power-quality compliance test types end to end (4-11, 4-14, 4-27, 4-28, 4-29, 4-34), plus their Korean equivalents KS C 9610-4-11 and KS C 9610-4-29, from the standard to lab validation, including the USB communication and calibration of the voltage-dip generator.
- Made an unreliable legacy instrument dependable by redesigning how the software communicates with it: tests no longer stall and now meet the timing the standards require.
- Migrated two complete legacy C# test applications into the new platform, reconstructing requirements through source-code analysis, communication traces, and reverse engineering, and improving their logic and functionality.
- Developed the reporting system (editable templates, DOCX/PDF/RTF generation) and the serialization and configuration-management infrastructure, with versioned data migrations that keep user data compatible across releases.
- Managed dependencies and builds with Conan, CMake, Git, and npm in a Kanban workflow (Jira), and introduced feature flags to enable or disable functionality per release.
- Worked with an international team of engineers and testers on a product used by test labs in the US, Europe, and China.

ELECTRONICS ENGINEERING INTERN — **VZ INGENIERÍA** — **Mar 2022 – Aug 2022**
Research and Development Team — Paraná, Argentina

Autonomous waste-sorting bin prototype

- Developed embedded firmware in C/Assembly for PIC microcontrollers, implementing sensor reading and actuator control logic.
- Designed the electrical schematics and PCB layout using Altium Designer.
- Built and brought up the prototype, performed functional testing and debugging, and produced its technical documentation: schematics, BOM, flow diagrams, and cost estimation.

Electric thermostat installation

- Produced the wiring analysis and the BOM, documented in a technical report.

PROJECTS

Personal Portfolio · frndrn.github.io

- Built a bilingual (EN/ES) portfolio in Angular using standalone components, signals, zoneless change detection, and Angular Material, with responsive layout, light/dark themes, and static prerendering; deployed to GitHub Pages.

Intelligent Wildlife Monitoring System (Final Degree Project)

- Designed and built a solar-powered camera trap integrating Raspberry Pi, Arduino Nano, remote access, night vision, and automated image cataloging.
- Developed a PyQt-based desktop application (UI designed in Qt Creator) for configuration, monitoring, and control of the system.
- Implemented a 360° detection and positioning system using PIR sensors and a stepper motor.
- Designed electronic circuits and PCB in KiCad for power management, environmental sensing, positioning, and thermal control.
- Created and curated image datasets and trained transfer learning models based on VGG16 and VGG19 for automatic animal classification.

Report: https://ria.utn.edu.ar/handle/20.500.12272/8159

TECHNICAL SKILLS

- **Programming Languages:** C/C++, TypeScript, Python
- **Frontend:** Angular, RxJS, Signals, Angular Material, Chart.js, Electron, HTML/CSS
- **Backend:** REST APIs (Crow), API design, multithreading, Boost
- **Tools:** Git, CMake, Conan, npm, Qt Creator, ESLint, Prettier, VS Code, Visual Studio, Jira, Linux
- **Data Formats:** JSON, XML
- **Testing & Validation:** automated testing, functional validation, compliance testing
- **Software Lifecycle:** versioning, configuration management, serialization
- **Hardware & Embedded:** USB (libusb), SCPI, TCP/IP, Serial, mDNS, MCC DAQ, PIC, AVR, UART/I2C/SPI, Raspberry Pi, Arduino, KiCad, Altium Designer

EDUCATION

- **Electronics Engineering** (6-year degree), National Technological University (UTN), Argentina **2017 – 2023**

SOFT SKILLS

Ownership and autonomy, clear written communication with remote international teams, cross-functional collaboration (software, hardware, and testing), analytical problem solving, attention to detail, and self-directed learning.

LANGUAGES

Spanish (Native), English (B2).